Inventory and Supply Chain Systems
EIN4343 Section 6493
Spring 2026

6493:26291: MWF Period 6 (9:35 PM - 10:25 PM) Larsen Hall 330

Instructor
Dr. Elif Akçalı. akcali@ise.ufl.edu 352-294-7726. Weil 450A. Office Hours: M 10:45AM-12:00PM or by appointment

Teaching Assistant
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Course Description
Deterministic and stochastic inventory models for single- and multiple-item systems. Analysis and design of logistics systems. Supply chain management and coordination. Demand forecasting.

Course Pre-Requisites / Co-Requisites
ESI3312 (Operations Research 1) with a minimum grade of C.

Course Objectives
This course develops student's analytic abilities to formulate and solve inventory, logistics and supply chain problems faced by today's firms. Students learn to take a comprehensive view of complex inventory and supply chain systems. Additionally, the students develop abilities to model, optimize, and design such systems.

Materials and Supply Fees
N/A

Professional Component (ABET)
This course contributes to the following PEOs:
1. Can become successful professionals in industrial and systems engineering or other disciplines
2. Can acquire advanced knowledge through continuing education or advanced degree programs

Relation to Program Outcomes (ABET)

Outcome	Coverage*
1. An ability to identify, formulate, and solve engineering problems by applying principles of engineering, science, and mathematics.	High
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	Medium
3. An ability to communicate effectively with a range of audiences.	Medium
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	Low
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	Low
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	Medium

*Coverage is given as high, medium, or low. An empty box indicates that this outcome is not covered or assessed in the course.

Required Textbooks and Software

- Lecture Notes
  - Title: EIN4343 Lecture Notes
  - Author: Elif Akçalı
  - Available at Course Site
- Supplemental Reading
  - Throughout the semester several essays/articles from contemporary business literature will be posted on the course website that you will be asked to read for follow-up class discussions.

Recommended Materials

Please get a subscription for The New York Times and Wall Street Journal through one.ufl.

Course Schedule

This is a tentative outline. The instructor reserves the right to make changes as she sees necessary.

Week	Date	Topic	Activity
1	1/12	Introduction: Syllabus and Motivation	
	1/14	Inventory Management: Preliminaries	
	1/16	Inventory Management: ABC Analysis	
2	1/19	**Martin Luther King Day** *No Class Meeting*	
	1/21	Inventory Management: Deterministic Models Economic Order Quantity (EOQ)	
	1/23	Inventory Management: Deterministic Models EOQ Model	I am Poem Homework 1
3	1/26	Inventory Management: Deterministic Models EOQ Model Extensions I	
	1/28	Inventory Management: Deterministic Models EOQ Model Extensions II	In Class Activity 1
	1/30	Inventory Management: Stochastic Models Periodic Review: Single-period model (cost min)	Homework 2
4	2/2	Inventory Management: Stochastic Models Periodic Review: Single-period model (profit max)	
	2/4	**UF Career Showcase** *No Class Meeting*	
	2/6	Inventory Management: Stochastic Models Periodic Review: Single-period model	Technical Poem 1 In Class Activity 2
5	2/9	Inventory Management: Stochastic Models Periodic Review: Multi-period models	
	2/11	**Term Exam 1** *In-Class/Evening Exam (TBD)*	Term Exam 1
	2/13	Inventory Management: Stochastic Models Periodic Review: Multi-period models	Price Research
6	2/16	Inventory Management: Stochastic Models Continuous Review: (q,r) model	Homework 3
	2/18	Inventory Management: Stochastic Models Continuous Review: (q,r) model	In Class Activity 3
	2/20	Inventory Management: Stochastic Models Continuous Review: (q,r) model	
7	2/23	Inventory Management: Stochastic Models Continuous Review: (q,r) model	
	2/25	Inventory Management: Stochastic Models Continuous Review: Service Levels (Type I)	

	2/27	Inventory Management: Stochastic Models Continuous Review: Service Levels (Type II)	
8	3/2	Transportation Management Preliminaries	Homework 4
	3/4	Logistics Management Preliminaries	
	3/6	Logistics Management Shipment Mode Selection	Technical Poem 2: Draft
9	3/9	Logistics Management Shipment Size Selection	
	3/11	Logistics Management Demand Pooling	In Class Activity 4
	3/13	Logistics Management Network Design	SC Analysis
10	3/16	**Spring Break** *No Class Meeting*	
	3/18	**Spring Break** *No Class Meeting*	
	3/20	**Spring Break** *No Class Meeting*	
11	3/23	Logistics Management Consolidation I	
	3/25	**Term Exam 2** *In-Class*	
	3/27	Logistics Management Consolidation II	
12	3/30	Logistics Management International Considerations	Homework 5
	4/1	Supply Chain Management Definition and Fundamental Issues	
	4/3	Supply Chain Management The Bullwhip Effect	
13	4/6	Supply Chain Management Integration	Supply Chain Analysis
	4/8	Supply Chain Management Contemporary Issues	In Class Activity 5
	4/10	Demand Forecasting Preliminaries and Forecasting Error	
14	4/13	Demand Forecasting Methods for Stationary Series	Homework 6
	4/15	Demand Forecasting Methods for Stationary Series	
	4/17	Demand Forecasting Methods for Linear Trend Series	
15	4/20	Demand Forecasting Methods for Linear Trend Series	
	4/22	Demand Forecasting Methods for Linear Trend Series	In Class Activity 6
	4/24	Demand Forecasting Methods for Seasonal Series	Technical Poem 2: Revised
16	4/22	Demand Forecasting Methods for Seasonal Series	
	4/24	Demand Forecasting Contemporary Issues	Creative Thinking Skills Reflection
	4/26	**Reading Days** *No Class Meeting*	
17		**Final Exam** *WEIM 1084: 7455: 23816: April 29, 3:00pm-5:00pm*	Final Exam

Attendance Policy, Class Expectations, and Make-Up Policy

- **Class.** Students will be provided with detailed class notes. Students are expected to read through the class notes before coming to class. Classroom meetings will be used to (i) reinforce and ensure student learning; (ii) highlight distinguishing features of different decision-making tools and methodologies; (iii) work through example problems to enhance student learning; and (iv) discuss issues related to the use of the decision-making tools and methodologies for analysis in practice.
- **Re-grade Requests.** Requests for re-grading of any assignments and exams will be considered only with a written explanation and only within a one-week period from the time graded work is returned in class. Specifically, you need to prepare a typed-up or legibly written memo explaining why you believe your work needs to be regraded, attach make it the cover sheet of the exam booklet, and return it to the instructor. Grades will be posted on the course page on Canvas. Throughout the semester, you have to ensure that your grades are entered correctly. Corrections to posted grades will be considered only within one week after the grades have been posted on Canvas.
- **Announcements and Course Materials.** You are responsible for all announcements made in class as well as some additional material that might be covered.
- **Professional Conduct.** Please remember to put your cellular telephones on silent before you come to class. Please come to class on time if attending in person. Late arrivals as well as walking out and in during the lectures disrupt the flow of the class. *Persistent disruptive behavior will not be tolerated and might result in grade deductions (half a letter grade).*
- **Absences.** Excused absences must be consistent with university policies in the undergraduate catalog (https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx) and require appropriate documentation.

Exams, Assignments, Quizzes, and Activities.

- **Exams.** Students will be given two Term Exams and a comprehensive Final Exam. Formula sheets that can be used during the exams will be provided by the instructor. These formula sheets will be posted on the course website in advance. The dates of the exams are announced, please plan accordingly.
- **Homework Assignments.** Throughout the semester, there will be 6 homework assignments. You need to submit 4 throughout the semester.
- **In-class Activities.** There will be 6 in-class activities to support the integration of creative thinking and technical course content as well as the real-life issues and technical course content..
- **Supply Chain Analysis.** There will be three assignments throughout the semester that will ask you do think about the application of what you learn in class to broader supply chain issues. For instance, you will be asked to listen to a podcast about the operations of a business and respond to several questions to analyze and conceptualize the supply chain operations of the business. More information will be provided via Canvas.
- **Creative Thinking Skills Development.** The course focuses on the development of creative thinking skills of the students. You will be given creative writing and visual storytelling assignments throughout the semester.
  1. *I am Poem*. Write an "I am" poem about yourself.

     I am (two special characteristics)
     I wonder (something you are actually curious about)
     I hear (an imaginary sound)
     I see (an imaginary sight)
     I want (an actual desire)
     I am (the first line of the poem restated)
     I pretend (something you pretend to do)
     I feel (a feeling about something imaginary)
     I touch (an imaginary touch)
     I worry (something that really bothers you)
     I cry (something that makes you very sad)
     I am (the first line of the poem repeated)
     I understand (something you know is true)
     I say (something you believe in)

I dream (something you actually dream about)
I try (something you make an effort to do)
I hope (something you actually hope for)
I am (the first line of the poem repeated)

2. *Technical Poem 1.* You will be asked to write an "I am" poem on a topic, concept, model, issue, etc. related to deterministic inventory management. More information will provided via Canvas.
3. *Technical Poem 2: Draft*. You will be asked to write an "I am" poem on a topic, concept, model, issue, etc. related to stochastic inventory management. More information will provided via Canvas.
4. *Creative Response Process.* You will be asked to use the Creative Response Process (Lerman and Borstel 2003) to give and receive feedback in peer groups. More information will be provided via Canvas.
5. *Technical Poem 2: Revised.* You will be asked to revise your "I am" poem on a topic, concept, model, issue, etc. related to stochastic inventory management based on the feedback you receive from your peer group. More information will provided via Canvas.
6. *Creative Thinking Skill Development Reflection.* Respond to several questions focusing on your experience with the creative thinking skills development assignments. More information will be provided via Canvas.

Make-up Policy

- **Exams**. Make-up tests will be given for unanticipated absences ONLY (e.g., illness/death in family with appropriate documentation, but NOT for job interviews). Exam Policy is non-negotiable.
- **In-Class Activities***.* Make-up for in-class activities will not be provided. Lowest two scores will be dropped.
- **Homework and Other Assignments**. Late submissions up to 3 days will be accepted with 1-point deduction for each day late.

Evaluation of Grades

Your final grade for the class will be calculated as shown in the table below:

Term Exam 1	22%
Term Exam 2	22%
Final Exam	26%
Homework Assignments	8%
Creative Thinking	8%
Supply Chain Research and Analysis	6%
In-Class Activities	8%
Total	**100%**

Grading Policy

Final grade averages will be rounded off to the nearest integer, and assigned on the standard scale as follows:

A	**A-**	**B+**	**B**	**B-**	**C+**	**C**	**C-**	**D+**	**D**	**D-**	**E**
100-93	92-90	89-87	86-83	82-80	79-77	76-73	72-70	69-67	66-63	62-60	59-0

More information on UF grading policy may be found at:
https://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

Students Requiring Accommodations

Students with disabilities requesting accommodations should first register with the Disability Resource Center (352-392-8565, https://www.dso.ufl.edu/drc) by providing appropriate documentation. Once registered, students will receive an accommodation letter which must be presented to the instructor when requesting accommodation. Students with disabilities should follow this procedure as early as possible in the semester.

Course Evaluation

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/.

University Honesty Policy

UF students are bound by The Honor Pledge which states, "We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: "On my honor, I have neither given nor received unauthorized aid in completing this assignment." The Honor Code (https://sccr.dso.ufl.edu/policies/student-honor-code-student-conduct-code/) specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. If you have any questions or concerns, please consult with the instructor and/or Tas in this class. *All cases will be handed over to the Dean of Students.*

Software Use

All faculty, staff, and students of the University are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against University policies and rules, disciplinary action will be taken as appropriate. We, the members of the University of Florida community, pledge to uphold ourselves and our peers to the highest standards of honesty and integrity.

Student Privacy

There are federal laws protecting your privacy with regards to grades earned in courses and on individual assignments. For more information, please see: https://registrar.ufl.edu/ferpa.html

Academic Policies and Resources.

Please go to https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/